



Improvement of real-scale raceway bioreactors for microalgae production using Computational Fluid Dynamics (CFD)

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ABSTRACT

In this work, the design and hydrodynamic of a 500 m² raceway bioreactor were optimized using Computational Fluid Dynamics (CFD). First, the influence of the bend type (traditional, isle partition and baffle partition) and the liquid velocity was studied, after that the dynamic behavior of the optimal configuration was analyzed. A monophasic analysis employing the Finite Element Method (FEM) with COMSOL Multiphysics™ allows to confirm that the utilization of deflectors in the baffle partition bend type provide the best performance in terms of fluid velocity, reduction of dead zones, shorter residence time and a suitable cell Reynolds number. A multiphase analysis using the Finite Volume Method (FVM) with ANSYS Fluent was performed to complete the analysis, it considering the geometry and rotation speed of the paddlewheel. Different angular velocities from 14 to 20 rpm were studied. Results shows that at 18 rpm average liquid velocity of 0.38 m·s⁻¹ were achieved and stabilized at 240 s, the pressure drop in the overall system stabilizes at 700 Pa at 330 s, power consumption in the range of 30 W·m⁻³, a maximum turbulence intensity of 0.4 and vorticity greater than 9 s⁻¹ in areas adjacent to the paddlewheel being determined. In addition, it was determined that more than 14% of the total volume is useful for mass transfer with the atmosphere as well as to water loss by evaporation. This study provides a robust start point for improving large-scale raceway reactors, a highly relevant issue because these are the most used reactors in large-scale microalgae production.

1. Introduction

Microalgae have aroused great interest worldwide due to their extensive potential in the pharmaceutical and nutraceutical industries, in addition to renewable energy among others [22]. Raceway reactors are the most widely used production technology; in fact, more than 90% of systems worldwide are based on these types of reactors. One of the reasons for that is because raceways can easily be scaled up, allowing high productivity at low energy costs and requirements although they are only suitable to produce roust strains resistant to contaminations [25]. The design of raceway reactor has a significant impact on biomass production rates because it influences relevant variables such as the degree of mixing, then influencing the existence of gradients of properties such as temperature, CO₂ availability, dissolved oxygen concentration and pH, but specially light penetration and their use by the microalgae cells [1]. Thus, raceway reactor design and operational

conditions have a large impact into the adequacy of culture conditions, then on biomass productivity, in addition to into the power consumption and overall performance by affecting both mass transfer and fluid-dynamic behavior [15,24,26,29,31].

To optimize the design and operational conditions of raceway reactors, different factors must be considered. Geometry of the reactor, configuration of bends, and specific design of paddlewheel being some of the most relevant ones. Geometry includes the volume to surface ratio (water depth) and aspect ratio (length to wide ratio). Regarding the volume to surface ratio, a typical value of 0.25 to 0.30 m³/m², corresponding to culture depth of 0.25–0.30 m, was proposed more than 40 years ago [28], although much larger water depth up to 1.0 have been proposed [32]. However, to improve the light-penetration it is recommendable to reduce the volume to surface ratio below 0.2 m³/m², culture depths up to 0.15 m demonstrating high yields. Nevertheless, shallow raceway bioreactor depths generate a series of fluid movement

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Fig. 1. Images of the real raceway reactor utilized, it being already installed at Las Palmerillas Experimental Station, Almería. Spain.

problems because friction losses increase and the hydraulic jump product of the rotary machinery (the paddlewheel) decreases, thus causing microalgal decay and the consecutive death of the microorganism. To reduce the water depth below 10 cm the utilization of thin-layer reactors is recommendable, it allowing to increase the biomass productivity by enhancing the light exposition of the cells [7].

Regarding aspect ratio an optimal value of 10 have been defined for large scale raceways. The higher is the aspect ratio the larger is the effect of walls of the reactor in the head losses then increasing the energy consumption. However, reducing the aspect ratio more dead zones, on which high-pressure and low-velocity vortices ($<0.1 \text{ m}\cdot\text{s}^{-1}$) exist. Both, water depth and aspect ratio determine the mixing on the reactor both longitudinal and transversal/vertical. To avoid a low level of mixing and thermal stratification, speeds above $0.05 \text{ m}\cdot\text{s}^{-1}$ are necessary [5,14], whereas to avoid sedimentation, velocities greater than $0.1 \text{ m}\cdot\text{s}^{-1}$ are required. Consequently, velocities greater than $0.2 \text{ m}\cdot\text{s}^{-1}$ are used in the straight sections [14].

The installation of baffles has been reported to improve mixing cycles in research looking at enhanced productivity [24,35]. Configuration of bends have a large effect into the energy consumption whereas its influence on mixing is quite low at large scale [24,26,35]. Usually the island configuration is recommended but by installing optimized baffles a better performance of the system can be achieved, definitively the use of not adequate bends largely reducing the performance of whatever raceway, including the complete fail of the system [24]. Regarding impulsion of the liquid, the utilization of paddlewheels is the most extended strategy, although also the utilization of airlift system and propellers have been proposed, but none of them being performed at industrial scale [12,21,37]. The design of paddlewheel systems have been already defined time ago, the utilization of paddlewheel with total diameter four times the water depth and a number of paddles from 10 to 12 being recommendable [36]. The efficacy of paddlewheel thus designed ranged from 20 to 50% according to the final quality of construction, the utilization of much more efficient propellers demonstrating efficiencies much higher, close to 90% but at much higher cost, then they are only recommendable for extremely low energy demanding processes such as wastewater treatment [9].

The optimization of raceway reactor designs and their configuration has been widely reported using oversimplified approaches with classical equations such as Manning or Bernoulli, or including experimental measurements [12,26,35]. These types of approaches are useful and suitable when considering a non-dependent space formulation as they only allow bulk water velocity estimations in simple geometries. Unfortunately, these optimization methods do not intrinsically consider the parts of the reactors overall, such as the bends and baffles, and require the building of the reactor to accurately evaluate them. Computational Fluid Dynamic (CFD) tools offer an alternative to circumvent these

obstacles. CFD allows to simulate the behavior of whatever raceway reactor, it being especially recommendable to optimize its design in terms of geometry of the reactor, configuration of bends, and specific design of paddlewheel.

Conventionally single phase systems are used on which a speed-input at the beginning of the reactor is considered, the software allowing to determine the pressure drop, liquid velocity, mixing and other fluid-dynamic variables along the reactor [5,18,23,24,34]. However, the paddlewheel imposes the existence of pulses on the movement of the liquid that influences the mixing and overall performance of the reactor [3]. To simulate this pulse related phenomena and accurately simulate the behavior of the system much more multiphase systems must be considered. Thus, the overall performance of raceways is greatly influenced by the phenomena taking place into the channel of the reactor but also by those taking place into the paddlewheel. Proper paddlewheel speed largely increases the gas exchange which mainly takes place around the paddlewheel since there is very low liquid/air interaction in the rest of the channel [14]. Otherwise, a very low angular velocity will not give the culture the necessary energy to mix whereas a very high speed can be harmful due to excessive friction - the microalgae tend to be very delicate, especially the filamentous ones. The areas adjacent to the paddlewheel are those favored by most culture parameters [3,26]. An appropriate turbulence intensity percentage must be achieved by the paddlewheel to create pressure and speed changes that generate controlled vertical vorticity [10]. If horizontal vorticity is generated, this can cause sediment, especially when accompanied by low turbulence intensity and uncontrolled absolute pressure – a problem affecting the low section of the photobioreactor. Other phenomena associated with gas exchange appear on the surface of the culture resulting from induced agitation. If this is not constantly controlled, evaporation phenomena might also occur that concentrate the salts contained and lead to saturation. Inadequate evaporation control can increase temperatures to levels that are not viable for microalgal growth [2].

The objective of this study is to optimize the design and operation of a 500 m^2 real-scale raceway bioreactor using CFD, differently from previous works analysing small scale systems. As a first step, a more conventional monophasic approach is employed to define the optimal design of the reactor. Subsequently, a more innovative and accurate but complex multiphase approach is used to evaluate the overall performance of the selected optimal design and to determine the influence of the operational conditions on the relevant hydrodynamic factors. This work uses CFD to optimize the design and operation of real scale raceways and promotes its use in the industrial application of this technology. Therefore, this work represents a relevant contribution in the design of large-scale raceway systems for microalgae production, essential for the expansion of this industrial activity.

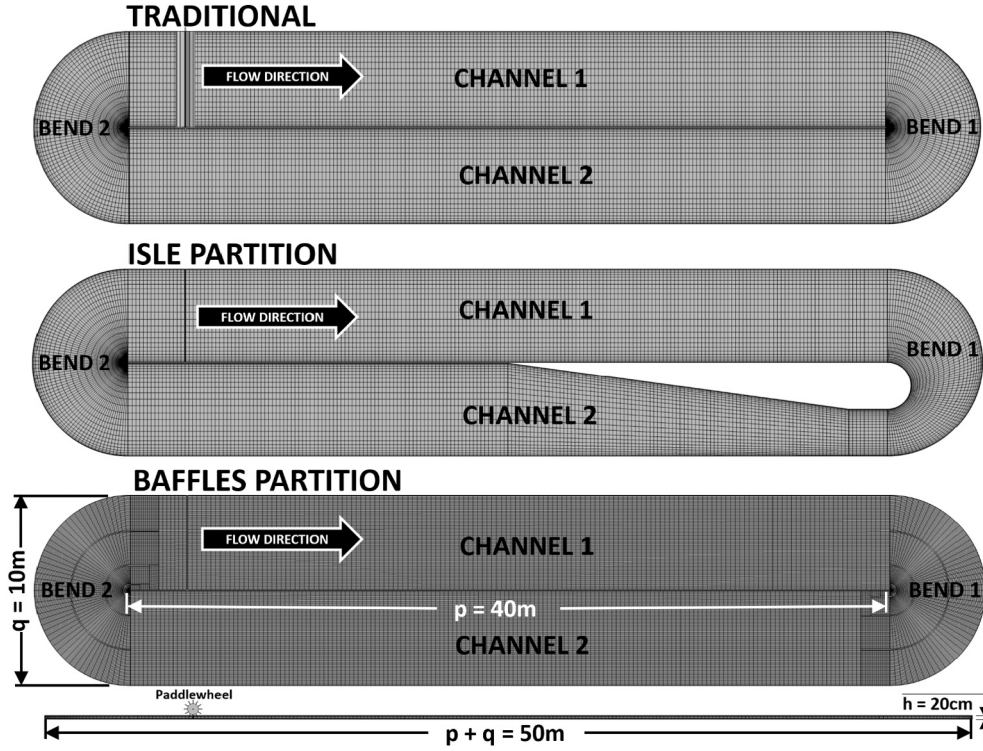


Fig. 2. Discretization domains of the three different configurations already considered obtained using COMSOL Meshing.

2. Materials and methods

2.1. Location of the real-scale raceway reactor

A 500 m² (10 × 50 m) raceway reactor was built at Las Palmerillas Experimental Station of the Fundación CAJAMAR (36°48'N–2°43'W), Almería, Spain. The reactor was operated at 0.2 m water depth although other water depths are possible, walls of the reactor having a total height of 0.4 m (Fig. 1). The reactor was built using a 1.5 mm polyethylene sheet arranged into the ground directly, walls being conformed using stainless steel stink located at 1.0 m distance. The real configuration of the reactor was used to define the geometry of the simulated reactor. The reactor is fully automatized, including the paddlewheel's rotatory velocity, the water depth inside the reactor. A stainless-steel paddlewheel of 1.0 m diameter with 10 paddles was utilized. The velocity of the paddlewheel was controlled by an inverter connected to an engine of 1.5 kW (Ebara, Barcelona, Spain). Bends of the reactor were designed according to optimal values found in this paper.

2.2. Design phases with computational fluid dynamics

Two different software tools were used: (i) COMSOL Multiphysics™, which uses the Finite Element Method (FEM), and (ii) ANSYS-Fluent CFD Research, which uses the Finite Volume Method (FVM). COMSOL was used separately in three case studies to evaluate the most widely used raceway bioreactor configurations (geometries) in monophasic flow. Single-phase flow approaches are widely used due to their low computational cost. Fluent was used to evaluate the behavior of the multiphase flow photobioreactor (the air-water interaction) more accurately as well as the rotary machine (the paddlewheel) as a generator of agitation, so as to more accurately assess the reactor's operational behavior.

2.3. First modelling design step. Discretization domains and governing equations

In CFD analysis, it is difficult and complex to treat a rotating geometry solid domain in two phases (air and water) simultaneously submerged in the water for the COMSOL code [3]. Therefore, the rotary system and the surface air layer comprised the only hydrodynamic analysis domain for the water domain. Such a method is valid in this type of analysis and has been adopted by numerous researchers in the past [3,5,18,24,34]. Three different initial reactor configurations were considered in 3D (Fig. 2). The meshes were created with the COMSOL Multiphysics mesh generator, hexahedrally structured with a 97–99% symmetrical quality and an average element size of 0.2 L. The number of elements were: TRADITIONAL = 100,000, ISLE PARTITION = 111,800 and BAFFLES PARTITION = 364,000. Newtonian fluid water with a density of $\rho = 1000 \text{ kg}\cdot\text{m}^{-3}$ and a viscosity $\mu = 0.001 \text{ Pa}\cdot\text{s}$ was used as the initial condition. A regular flow velocity $u = 0.22 \text{ m}\cdot\text{s}^{-1}$ was set and on the top of the raceway, an atmospheric pressure was defined. The boundary walls had an average roughness of $Ra = 0.0015 \text{ mm}$.

Fundamental equations governing the monophasic flow problem at steady state are as follows:

$$\rho(u \cdot \nabla)u = \nabla \cdot [-pl + K] + F + \rho g \quad \text{Eq. (1)}$$

$$\rho \nabla \cdot (u) = 0 \quad \text{Eq. (2)}$$

$$K = (u + u_T)(\nabla u + (\nabla u)^T) \quad \text{Eq. (3)}$$

where F = volume forced vector ($\text{N}\cdot\text{m}^{-3}$), $T = T_{\text{ref}} = 293.15 \text{ K}$, $p = p_{\text{ref}} = 101,130 \text{ Pa}$, $g = 9.81 \text{ m}\cdot\text{s}^{-2}$, $l = 100 \text{ m}$. In this study the standard k - ε turbulence model was utilized [35]:

$$\rho(u \cdot \nabla)k = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_k} \right) \nabla k \right] + P_k - \rho \varepsilon \quad \text{Eq. (4)}$$

$$\rho(u \cdot \nabla)\varepsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_{\varepsilon 1} \frac{\varepsilon}{k} P_k - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k} \quad \text{Eq. (5)}$$

$$\mu_T = \rho C_\mu \frac{k^2}{\varepsilon} \quad \text{Eq. (6)}$$

$$P_k = \mu_T [\nabla u : (\nabla u + (\nabla u)^T)] \quad \text{Eq. (7)}$$

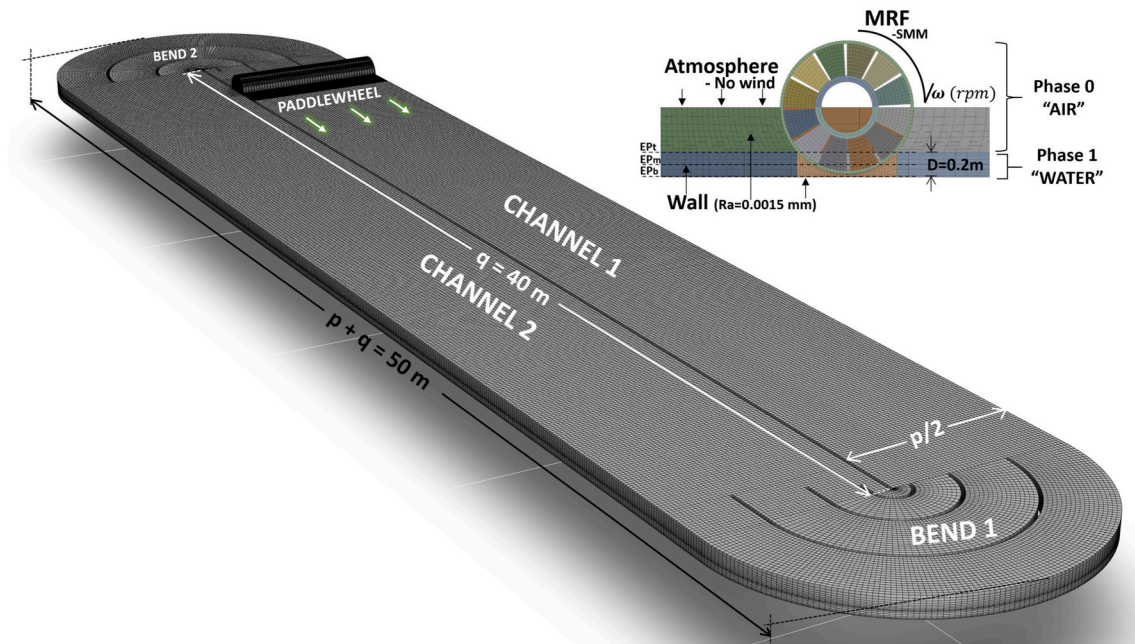


Fig. 3. Overall geometry and computational domain of the raceway reactor performed using ANSYS Meshing.

where, k = turbulent kinetic energy, ε = turbulent dissipation rate, μ_T = turbulent viscosity (pa-s), $\sigma_k = 1.0$ (Prandtl number for kinetic energy), $\sigma_\varepsilon = 1.3$ (Prandtl number for dissipation rate), $C_{\varepsilon 1} = 1.44$ and $C_{\varepsilon 2} = 1.92$ (first and second constant for the dissipation rate), and $C_\mu = 0.09$ (turbulent model constant).

The negative effect of the Dead Zone (DZ) is directly related to the fall in productivity energy loss, and others; velocities lower than $0.1 \text{ m} \cdot \text{s}^{-1}$ [14] will be considered dead volumes. These can be calculated using the following model [3]:

$$DZ = \frac{\text{Volume with velocity less than } 0.1 \text{ m/s}}{\text{Total volume of pond}} \quad \text{Eq. (8)}$$

The flow in a raceway needs to be turbulent to keep the cells in a mixing condition. Whether the flow is turbulent or not depends on its Reynolds Number, Re , which is defined as:

$$Re = \frac{\rho u \sqrt{d_c}}{\mu} \quad \text{Eq. (9)}$$

where, $\sqrt{d_c}$ is the equivalent diameter of an element, measured by meshing controls.

2.4. Second modelling design step. Discretization domains and governing equations

For this second CFD analysis, the geometrics were discretized using ANSYS Software Meshing (Fig. 3). The hexahedron meshing method was employed, achieving a total of 843,624 control volumes and 999,333 nodes with the linear order of elements. The FVM was used to predict the hydrodynamic behavior of the raceway in its transient regime. Navier-Stokes equations were used, which arise from Newton's Second Law of fluid motion. Considering that fluids such as water (which is incompressible) and air (which is compressible) were being calculated, two mass and momentum conservation equations were used, respectively.

To solve the multiphasic problem, a Eulerian approach was used in

which all the phases are calculated as a continuous medium. In this situation, the volume fractions are modelled through continuous functions in time and space throughout the domain. The Volume Of Fluid (VOF) technique was used [19], this being a technique developed for monitoring a surface between two or more fluids based on a fixed Eulerian mesh, where one region of the volume cannot be occupied by another. If α_q is considered the fraction of volume of fluid q , there are three possible cases: if $\alpha_q = 0$, the cell contains 0% of the fluid q (fluid = p); if $\alpha_q = 1$, the cell contains 100% of the fluid q whereas if $0 < \alpha_q < 1$, the cell contains the interface between the fluid and any other that is being considered. To solve the implicit scheme for temporal discretization, given that the equation requires knowing the volume fraction values in the current time step, a scalar transport equation is solved iteratively for each secondary phase volume fraction at each time step. This equation has the following form:

$$\frac{1}{\rho_q} \left[\frac{\partial}{\partial t} (\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) \right] = S_{\alpha_q} + \sum_{p=1}^n (\dot{m}_{pq} - \dot{m}_{qp}) \quad \text{Eq. (10)}$$

where $\dot{m}_{pq}$ and $\dot{m}_{qp}$ in the rate of mass transfer, S_{α_q} is zero but we can specify a constant or user-defined mass source for each phase.

The Reynolds-Averaged Navier Stokes Equations (RANS) was used with the Realizable $k - \varepsilon$ turbulence model [29]; this is the most recommended turbulence model when there is a paddlewheel with the angular velocity. In the transport equation for k , the constants (σ_k , σ_ε , $C_{\varepsilon 1}$, $C_{\varepsilon 2}$) are the same as in the standard $k - \varepsilon$ turbulence model but the local acceleration term must be added ($\rho \left(\frac{\partial k}{\partial t} \right)$). The modified realizable ε models are:

$$\frac{\partial(\rho \varepsilon)}{\partial t} + \frac{\partial(\rho \varepsilon v_i)}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_T}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_i} \right] + \rho C_{1\varepsilon} S_\varepsilon - \rho C_{2\varepsilon} \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} + C_{\varepsilon 1} \frac{\varepsilon}{k} C_{3\varepsilon} G_b S_\varepsilon \quad \text{Eq. (11)}$$

where μ_T = turbulent viscosity (not constant).

The raceway paddlewheel with a diameter of 1 m and 4.8 m wide is simulated and by a Multiple Rotating Reference Frame with 12 blades that interact with stationary domains. In this case, the Rotating Reference Frame (MRF) model, incorporating the Sliding Mesh Model (SMM)

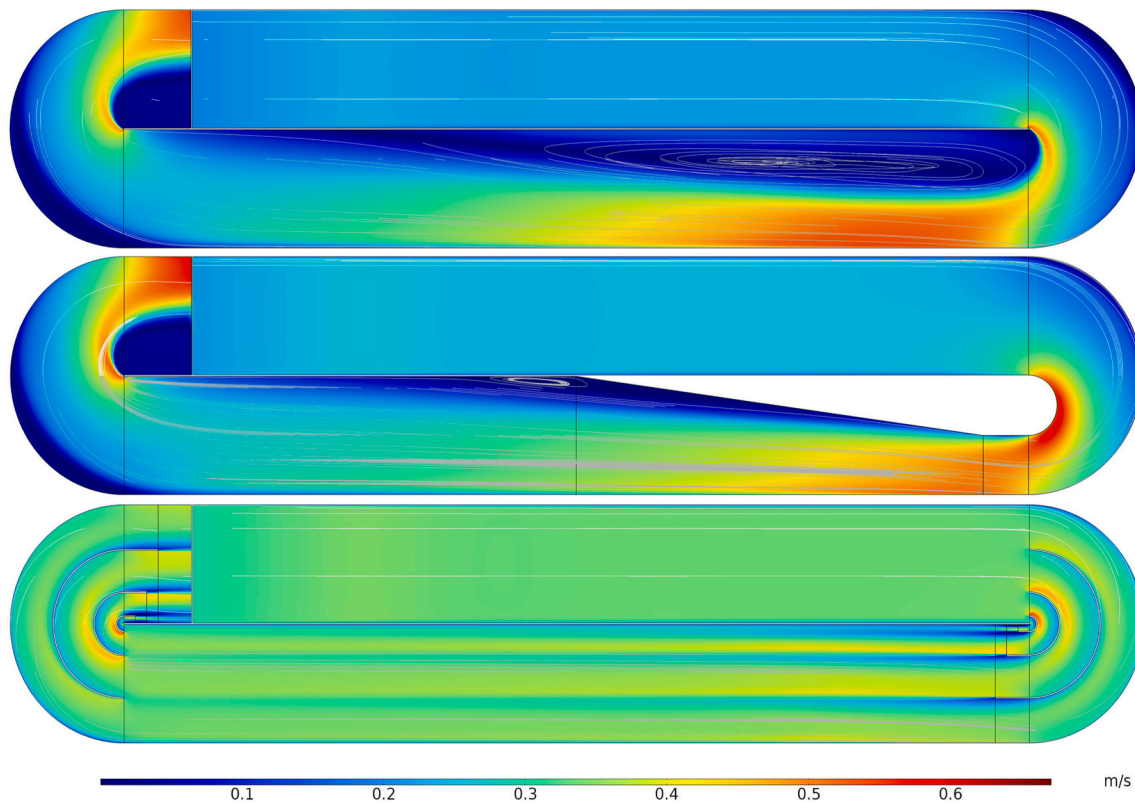


Fig. 4. Monophasic analysis of the raceway reactor: Velocity profiles. a) Traditional. b) Isle Partition. c) Baffles Partition.

[20] was used, this is composed of sliding meshes that capture the transient effects of movement in adjacent areas such as the paddlewheel, which moves at angular velocities of 14, 16, 18 and 20 rpm and interacts with Phase 0 and Phase 1. The transformed fluid velocities from the stationary frame to the rotating frame are:

$$\vec{v}_r = \vec{v} - (\vec{\omega} \cdot \vec{r}) \quad (12)$$

where $\vec{v}_r$ is the relative velocity, $\vec{v}$ is the absolute velocity, $\vec{\omega}$ is the angular velocity of the reference frame relative to the stationary reference frame and $\vec{r}$ is the distance of the stationary control volume from the rotating reference frame.

The effect of the inlet velocity and mixing is given by the paddlewheel while the regions of phases are given as initial conditions; these regions undergo variations as the VOF equations are solved. The wall boundary condition was used for the bottom and side walls with a roughness of 0.0015 mm while the top wall was a pressure-outlet. The paddlewheel blades were considered as a wall with no roughness. For the results visualization, three Elevation Plans were established in the liquid domain: 1) EP_t, on the surface, 2) EP_m, at half the depth (0.1 m) and 3) EP_b, on the floor at maximum depth.

The Pressure-Based Solver was used via the PISO Algorithm because, among its advantages, it works by correcting adjacent areas, such as the paddlewheel, with the rest of the domain, and higher time steps can be evaluated by decreasing the calculation time for large systems; although this involves a higher computational cost and decreases the chances of divergence.

3. Results and discussion

3.1. Monophasic analysis

Monophasic analysis allows a first approach to the system to evaluate the behavior of the three considered configurations and select the

optimal configuration for the raceway. On this analysis also the influence of liquid velocity on the reactor's overall performance is considered. Thus, one of the most important factors in microalgae reactors is the speed at which the fluid is recirculated along the reactors, because it influences the light exposition of the cells in addition to the existence of gradients of properties (pH, dissolved oxygen temperature, etc.) including nutrients which can reduce the performance of the cells. Various fluid dynamic parameters are governed by the liquid velocity - at low speeds, large vortices are generated that become dead zones, reducing the culture's productivity and efficiency; long residence times mean a high sedimentation capacity and low vertical rotation frequency of the microalgae culture. Having an adequate liquid velocity ensure enough turbulence so minimizing the existence of gradients of properties also increasing the light exposition of the cells.

The velocity profile and flow lines of the three configurations already considered is showed in Fig. 4. Data shows as the TRADITIONAL design has higher velocity gradients (particularly in the contour) than the other two designs; ultimately, this will mean a greater economic cost in mechanical energy and far more biological cellular stress. In addition, there is a dead zone after the first bend (Fig. 4). In the ISLE PARTITION scenario, the dead zone is clearly smaller than in the TRADITIONAL design. Nevertheless, the current's venturi choking effect increases the head loss and will evidently require greater mechanical energy. The BAFFLES PARTITION scenario is the most interesting because the gradient velocity, as well as the volume of dead zones, is the lowest. It can be observed that using baffles has the advantage of ordering the fluid currents in relation to the A/V so that they can rotate with the necessary mechanical energy and reduce the rebound effect caused by the bends. The adequacy of BAFFLES PARTITION configuration has been previously reported by other authors both from CFD and real measurements [24,26,35]. Moreover, the specific design of the bends must be optimized for each case. In the most conventional design the bends are half-circles of reducing radius from the outer wall, however in the design here proposed the baffles are longer at the outlet, to improve the

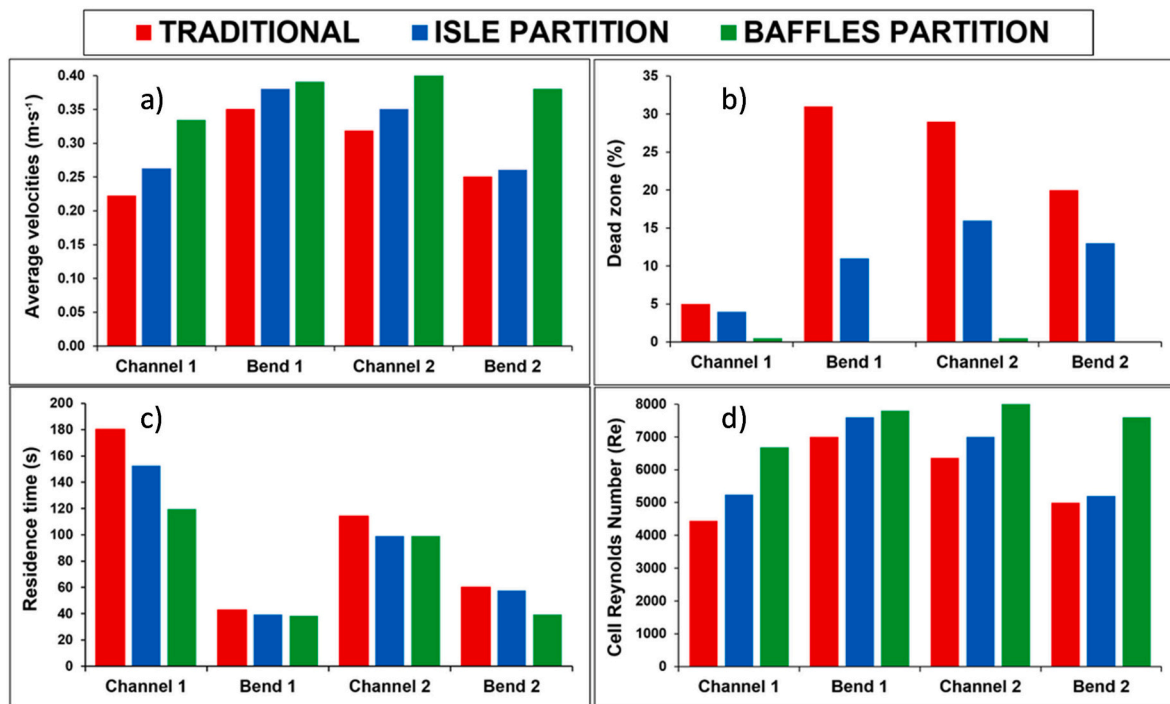


Fig. 5. Graph of monophasic study. a) Average velocities. b) Dead zones. c) Dispersion coefficient. d) Cell Reynolds Number.

ordering of the flow at the outlet of the bends, thus minimizing the creation of vortex, so the dissipation of energy at this position.

When analysing the overall performance of the three considered designs (Fig. 5a), it is observed as the BAFFLES PARTITION configuration provides the best results in comparison with the other two configurations. Regarding average velocity, the TRADITIONAL and ISLE PARTITION designs have a similar tendency in both channels, maintaining a velocity not exceeding $0.25 \text{ m} \cdot \text{s}^{-1}$ and producing a sharp velocity increase in Bend 1, mainly due to the choking effect produced by the high-pressure vortex and the isle type partition, respectively. In both cases, the vortex and the septum produce a similar hydrodynamic effect in Channel 2 and Bend 2, the velocity drops below $0.1 \text{ m} \cdot \text{s}^{-1}$ provoking high load losses and possible sedimentation [14]. The BAFFLES PARTITION design performs differently, there is an immediate velocity gain of $0.15 \text{ m} \cdot \text{s}^{-1}$ in Channel 1 and a slight velocity increase in Bend 1, which is energetically and biologically beneficial. In terms of the shear effect on the microalgae cells, the loss of velocity in Channel 2 and Bend 2 is practically negligible; the fluid transit remains constant while the turbulence effect constantly mixes the fluid, resulting in a low sedimentation capacity. The existence of large liquid velocity gradients in raceway reactors is a well-known phenomenon, the bottlenecks being to avoid the existence of zones with excessively low liquid velocities favouring sedimentation of the cells, such as already reported values of $0.05 \text{ m} \cdot \text{s}^{-1}$ [31].

Regarding the presence of dead zones, these have large disadvantages such as accumulation/precipitation of biomass and nutrients, the dissipation and loss of mechanical energy, all of them reducing the productivity of the cultures. In the TRADITIONAL design the presence of dead zones averages 22%, whereas in the ISLE PARTITION design the dead zones represent 10%, and in the BAFFLES PARTITION design the total of dead zones is less than 1% (Fig. 5b). By sections, the greatest dead zone presence in the TRADITIONAL design is found in Bend 1 and Channel 2, whereas in the BAFFLES PARTITION design there is no presence at all. Decreasing the dead zones greatly improves the reliability of raceway reactors by improving the performance of the microalgae cultures. The existence of large dead zones in the raceway reactors is a well-known problem for the stability of microalgae cultures.

On these zones the microalgae biomass can settle, then (i) it being not exposed to the light to perform photosynthesis then to produce more biomass, and (ii) it being easily corrupted favouring contamination problems. Previous works confirm that in the TRADITIONAL design dead zones can represent from 20 to 40% of the overall volume of the culture [24].

Regarding residence time, this variable provides a measurement of the movement of the fluid along the reactor, thus for a same surface and volume the lower the residence time the higher is the liquid velocity and homogeneity of movement of the fluid along the reactor. The residence times observed in Channel 1 are more than 150 s for the TRADITIONAL and ISLE PARTITION configurations due to the disturbance produced by the high pressure generated in the curves; this acts as an obstacle to fluid transit, producing greater retention of microalgae cells (Fig. 5c). As a result, liquid velocity is lower on this section which favored sedimentation, then inadequate exposition of the cells to light. The BAFFLES PARTITION configuration does not exceed 120 s in the channels and there is an orderly flow. In all the case studies, neither Channel 2 nor either of the curves would pose any problems to fluid transit.

Regarding turbulence, the cell Reynolds number in the different sections of the raceway reactor for the three designs already considered have been calculated (Fig. 5d). The challenge is to achieve a high turbulence to improve the mixing, specially the light exposition of the cells, while minimizing the energy consumption, always in a safe range for the cells avoiding damaging shear rates [13]. In the BAFFLES PARTITION design, the larger Reynolds number are achieved, it ranging from 8000 to 6800 at the channel 1 to 8000 at Channel 2. Opposite the TRADITIONAL design provides the lowest values ranging from 4000 to 7000 at channel 1 and bend 1 respectively. This is one of the most important criteria when analysing different reactors of operational conditions, it being recommended to maintain flow uniformity along the reactor at higher turbulence as possible [14]. Both the TRADITIONAL and the ISLE PARTITION configurations fail to reach the minimum safe criterion value of 8000, since the flow needs to be turbulent in order to ensure that microalgae are in suspension, to avoid thermal stratification and to favor vertical mixing, so the light exposition of the cells [14]. However, it has been reported that light/dark cycles at which the cells are exposed

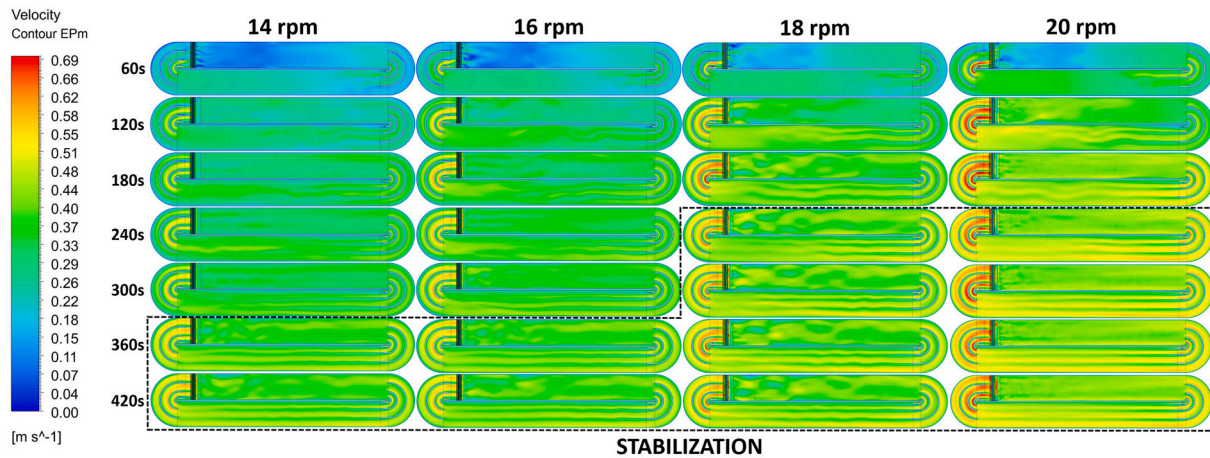


Fig. 6. Multiphasic analysis of the raceway reactor: Influence of rotation velocity of the paddlewheel on the stabilization time of the system.

to in raceway reactors is far from values allowing the light integration by the cells. Using CDF, it was reported that the light exposition times reduces from 3 to 1 s, whereas the dark exposition times reduces from 22 to 20 s, when the liquid velocity increases from 0.15 to $0.35 \text{ m}\cdot\text{s}^{-1}$ [10]. These values are far away from optimal frequencies of light exposition of 1 Hz [8]. Thus, in conventional raceway reactors it has been reported that to try to achieve full integration of light is impossible, or requires enormous amount of energy, in these reactors adaptation to local irradiance taking place [6]. The only way to approach to light integration in open reactors is to largely reduce the water depth, because the liquid velocity cannot be increased too much without damaging the cells or consuming large amounts of energy, then approaching to the thin-layer design. Although the design of thin-layer reactors was proposed time ago [16], the optimal design and fluid-dynamic on these reactors still must be optimized [33].

The results of the monophasic analysis clearly indicate that the BAFFLES PARTITION design meets all the important hydrodynamic conditions. Providing a reasonable potential for achieving successful microalgae production. In the next section, a more accurate analysis of this design is obtained by performing a multiphasic analysis.

3.2. Multiphasic analysis

A more accurate analysis of the baffles partition design was performed by multiphasic analysis, it being performed considering four different angular velocities for the paddlewheel. The Elevation Plane middle (EPm) considered is located at a 10 cm depth (Fig. 3). Multiphasic analysis allows to evaluate the variation of the system with the time, so its dynamic behavior. On this sense, the variation of liquid velocity profiles into the system with the time allows to determine the stabilization time, so the time required to achieve a constant behavior into the system. Data shows as the fluid velocity is stabilized after 360 s when using paddlewheel angular velocities of 14 and 16 rpm, and at 240 s for angular velocities of 18 and 20 rpm (Fig. 6). Once the system is stabilized, low velocities below $0.15 \text{ m}\cdot\text{s}^{-1}$ are observed when operating the system at 14 and rpm, then these rotational speeds being not recommendable [20]. Regarding liquid velocities at rotational speeds of 18 and 20 rpm, it is observed as flow is more chaotic and disordered in Channel 1 than in Channel 2 where the flow lines are controlled and smoothed with increased velocity. Difference in liquid velocity between inlet and outlet of the paddlewheel is due to the different water depth at these positions. Moreover, for a rotational speed of 20 rpm, suction resistance forces are generated which directly affect the paddlewheel due to the high-pressure difference. The minimum flow velocity identified in the boundary limits at the bottom ranged from $10^{-4} \text{ m}\cdot\text{s}^{-1}$ to $10^{-3} \text{ m}\cdot\text{s}^{-1}$ when the rotational speed of the paddlewheel modifies from

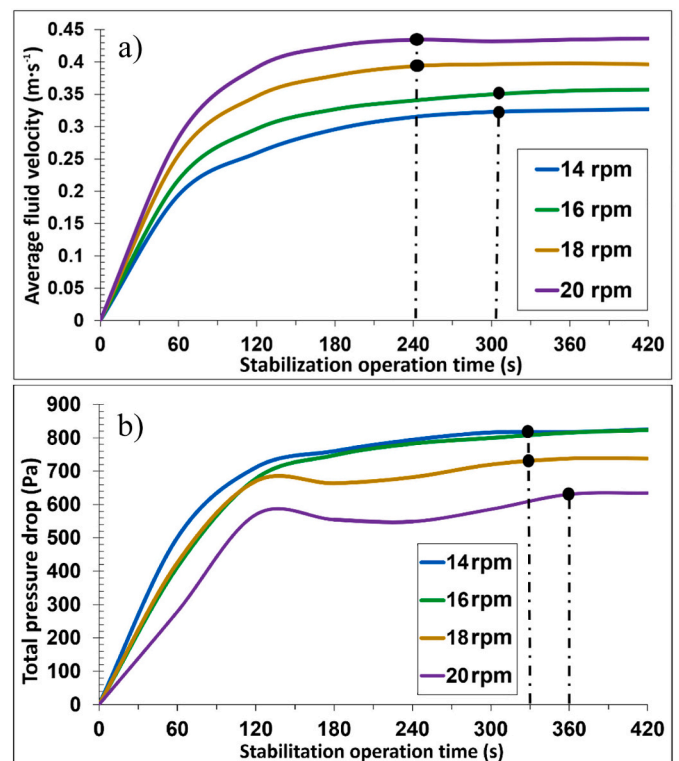


Fig. 7. Multiphasic study. Influence of rotation speed and time on the fluid-dynamic of raceway reactor. a) Average fluid velocity. b) Total pressure drop.

14 to 20 rpm. The highest liquid velocities are measured at water depths from 12 to 14 cm, with values that average around $0.4 \text{ m}\cdot\text{s}^{-1}$. Gradients of liquid velocity improve mixing between fluid layers allowing greater vertical particle-transit frequency. Otherwise, the existence of zones with extremely low liquid velocities provoke risk of sedimentation of microalgae cells on these zones, especially close to the bottom of the reactor (EPb). Moreover, there are other factors in addition to the fluid velocity that can favor sedimentation such as turbulence intensity, vorticity and, very importantly, relative pressure.

Average values of liquid velocity and pressure drop into the system at the different rotational speeds are summarized in (Fig. 7). Data shows as the higher the rotational speed the higher the liquid velocity, values ranging from 0.3 to $0.4 \text{ m}\cdot\text{s}^{-1}$ when the rotational velocity increases from 14 to 20 rpm (Fig. 7a). The rotational speed can be reduced to

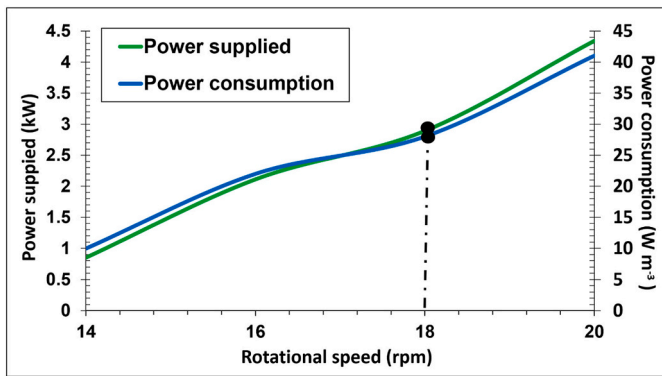


Fig. 8. Experimental data. Energy used for different operating conditions.

minimize the energy consumption but, in this case, low velocity zones favouring the sedimentation of the biomass, then the appearance of biological problems can occur. Opposite, to increase the liquid velocity in a reasonable range allows to increase the biomass productivity up to 30% [11]. The rotational speed is largely dependent of specific design of paddlewheel and overall reactor but it ranges from 5 to 10 rpm [10,19] to 30–40 rpm, although the most extended values ranges from 15 to 20 rpm [4,5,19,30,34]. Regarding the overall pressure drop into the system, data indicates that pressure drop reduces when increasing the rotational speed of the paddlewheel values reducing from 800 Pa at 14 rpm to 600 Pa at 20 rpm. This is because in spite of higher liquid velocity the flow is more ordered when operating at rotational speed of the paddlewheel of 20 rpm than at 14 rpm, then the dissipation of energy on the reactor reducing, this becoming more efficient. Fig. 8 shows the comparison between empirical and estimated energy consumption as a function rotational velocity of the paddlewheel. Both, empirical and simulated values provide similar results this confirming the adequacy of simulations already performed. Data shows as increasing the rotational velocity of the paddlewheel from 14 to 20 rpm the power required increases from 1 to 4 kW, specific power consumption also increasing from 10 to 40 $\text{W}\cdot\text{m}^{-3}$. At 18 rpm the power required is 3 kW specific power consumption being 30 $\text{W}\cdot\text{m}^{-3}$, at these conditions the liquid velocity being $0.36\text{ m}\cdot\text{s}^{-1}$. This values are much higher than previously reported values of $1.2\text{ W}\cdot\text{m}^{-3}$ [24], 2.2 to $5.0\text{ W}\cdot\text{m}^{-3}$ [19], but similar to values up to $20\text{ W}\cdot\text{m}^{-3}$ experimentally measured. [26]. Anyway, this study determined that the most balanced rotational velocity of the paddlewheel is 18 rpm although 20 rpm generates a little less pressure drop; nevertheless, with the latter, there is too much speed difference increasing the structural stress in Bend 2, the high velocity generated is

due to the high pulling power of the paddlewheel. The following studies are continued at 18 rpm and at the stabilization time of 240 s.

CFD simulations also allows to analyse the mass transfer phenomena, thus it being a powerful tool to have an early knowledge of the performance of different reactor designs or operational conditions. Fig. 9 shows the Volume Fraction between the water and air in the different sections of the reactor. The most interesting area is the interface corresponding to the fraction that is constantly mixing, the volume where the gas exchange balance with the environment is performed and where water is lost by evaporation. The area of greatest gas exchange is at the paddlewheel due to its agitation, and there are only a few centimetres in the channels and bends where the air is mixed (Table 1). Correct design in terms of mass transfer have a direct impact on productivity capacity, especially when considering oxygen removal which largely reduces the performance of large scale raceway reactors [27]. also, mass transfer into the channel is directly related with the carbon losses to the atmosphere through decarbonisation, although it has been already demonstrated that these are a minor contrition to the carbon balance on these systems if they are adequately operated [15,17]. Data also allows to

Table 1
Volume fraction summary.

Domain	Interface (%)	$V_{\text{domain}} (\text{m}^3)$	$V_{\text{interface}} (\text{m}^3)$
Paddlewheel	86	0.57	0.43
Water	14	99.43	13.92
Total		100	14.41

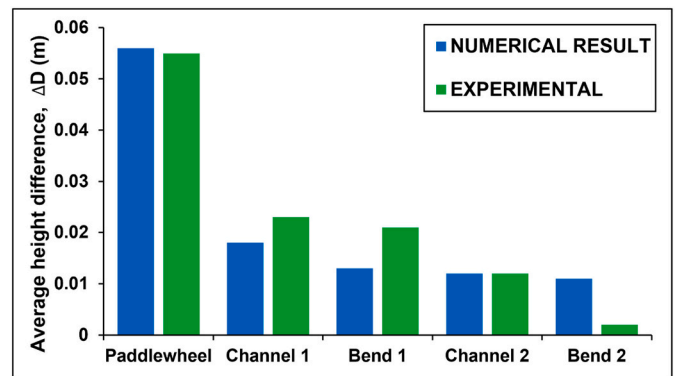


Fig. 10. Numerical and experimental comparison of culture height difference per section.

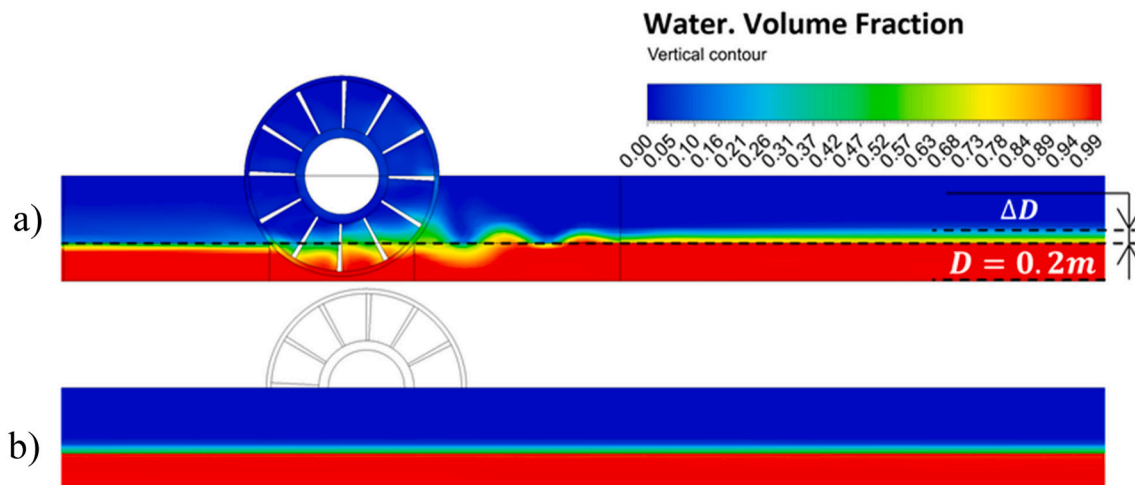


Fig. 9. Visual difference in the height of the culture column: a) Volume Fraction at Channel 1. b) Volume Fraction at Channel 2.

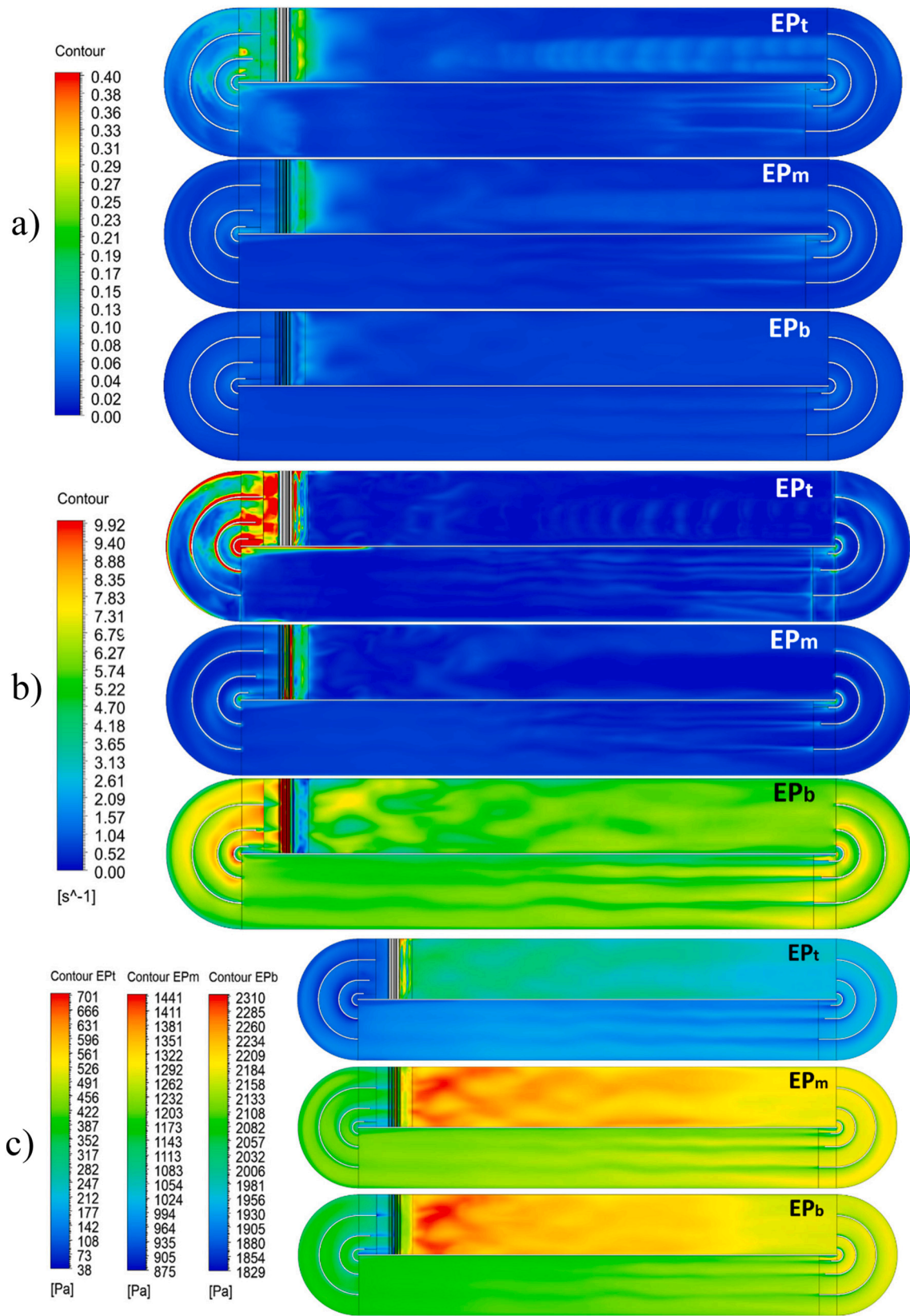


Fig. 11. Analysis contours: a) Turbulence intensity, b) Vorticity, c) Relative pressure.

observe the difference in height $\Delta \approx 0.05$ m due to the volume of fluid flowing after the mechanical drive, caused mainly by the increase in pressure in Channel 1 when encountering an obstacle (Bend 1). With regard to vorticity on the floor, this is a problem because it increases the absolute pressure favouring sedimentation in that lower part and a high total pressure being retained in Channel 1. Fig. 10 shows the experimental and simulated value of height difference in the different parts of the reactor. Both experimental and simulated values agree again confirming the adequacy of simulations already completed. Moreover, the pressure drop also agrees with the previous pressure drop determined into the system. Data shows as the culture depth decreases in the different sections, this different in water height inducing the circulation of the culture similar to what take place in a river achieving. Major difference in the water height was found at the paddlewheel, it being up to 0.055 m. The difference of water height is larger when the size of the reactor increases, it making difficult to scale-up raceway reactors at sizes larger than 5.000 m², especially if operating these reactors at average water depths below 0.3 m.

To conclude the analysis of the industrial scale raceway reactor the turbulence intensity, vorticity and relative total pressure into the system was calculated (Fig. 11). High turbulence intensity allows a reduction in the global drag forces which lead to the microalgae moving more freely. However, if it is low, the drag forces will stop them because they have generated increased friction in the boundary layer, permitting the appearance of tangential tensions if high vorticity is present in areas with less turbulence. Data shows as the turbulence slightly exceeds 0.2 at the places with the greatest vorticity due to the large pressure changes. Values above 0.1 indicates that they are highly disturbed causing a greater dissipation of cinematic energy. Regarding vorticity, it exceed 5 s⁻¹ due to the disorderly flow and turbulence generated by the paddlewheel; this is maintained in much of the Channel 1 section but dissipates when approaching Bend 2 and exceeds 8 s⁻¹ because of the drag forces of the rotary system that attracts the fluid increasing the tangential tensions; this is accompanied by high relative pressure. Consequently, the weight on the floor generates a retention of fluids that rotate and are finally retained in those areas. For this reason, studies should be carried out aimed at reducing the tangential tensions and the height differential caused by the paddlewheel. In EP_m and EP_t, it can be observed that the turbulence increases and the vorticities decrease less in Bend 2 while the relative pressure decreases, respectively. These are not considered to be major problems for the culture, but they should be improved in order to dissipate tangential tensions in Channel 1.

4. Conclusions

Monophasic analysis of different raceway reactor configurations allows to identify the deflector partition design as the most adequate for the production of microalgae, from the fluid-dynamic point of view. This design allows to minimize the possibility of biomass sedimentation by reducing the existence of dead zones and reducing the residence time in the different sections of the reactor, due to an optimal fluid movement on it. The turbulence at this configuration is also the optimal one, values of Reynold number upper than 6000 being determined all through the reactor. A further multiphasic analysis allows to perform a more accurate characterization of the baffle partition design, including at different rotational velocities of the paddlewheel. Data shows as multiphasic analysis provide accurate data from the reactor, experimental measurements of both energy consumption and differences in water height along the reactor validating simulated data. Simulations shows as proper fluid stabilization and liquid velocities were predicted at 18 rpm, at reasonable energy consumption. Data also confirm that in raceway reactors the volume available for mass transfer is low, existence of large differences in water height on the paddlewheel doing difficult the scale-up of this type of reactors at sizes larger than 5.000 m². CFD simulations allows to have a precise description of raceway reactors thus it being a powerful tool on the design and optimization of this type of reactors.

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Authors contributions statement

Cristina Inostroza perform most of the work, he being in charge of simulations and analysis of data. Alessandro Solimeno collaborate in the work already performed using COMSOL. Jose Maria Fernandez Sevilla and Joan Garcia collaborate in the analysis of data and writing the paper. F. Gabriel Acien also participate on the revision of data, discussion and preparation of the final documents.

Declaration of competing interest

The authors declare that there are any potential financial or other interests that could be perceived to influence the outcomes of the research.

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